Question 1

Simplify the following expression:

$$3(2x^3 - x^2 + 4 - 5x) - 2x^3 + x^2$$

A) $4x^3 - 2x^2 - 15x + 12$

B) $6x^3 - 2x^2 - 15x + 12$

C) $4x^3 - 4x^2 - 15x + 12$

D) $2x^3 - 4x^2 - 15x + 12$

WORKSPACE

Question 2

Consider the following formula. Which formula is equivalent to this one?

$$y = 3(x - 2)(x + 4)$$

A) $3x^2 + 6x - 24$

B) $3x^2 + 12x - 24$

C) $x^2 + 6x - 8$

D) $3x^2 - 6x - 24$

WORKSPACE

Question 3

Factor the polynomial. Which is the correct factorization?

$4x^3y + 8x^2y^2$

A) $4x^2y(x + 2y)$

B) $8x^2y(x + y)$

C) $4xy(2x^2 + y)$

D) $4x^3y(1 + 2y)$

WORKSPACE

Question 4

Factor the polynomial. Which is the correct factorization?

$x^2 + 5x - 24$

A) $(x + 8)(x - 3)$

B) $(x + 6)(x - 4)$

C) $(x + 12)(x - 2)$

D) $(x - 6)(x + 4)$

WORKSPACE

Question 5

What is the next number in the following series of numbers?

3, 6, 15, 42, ?

A) 96

B) 123

C) 135

D) 153

WORKSPACE

Question 6

What is the next number in the following series of numbers?

4, 9, 7, 14, 12, 24, ?

A) 18

B) 19

C) 22

D) 23

WORKSPACE

Question 7

What is the next number in the following series of numbers?

5, 9, 19, 37, 75, ?

A) 141

B) 149

C) 151

D) 153

WORKSPACE

Question 8

What is the next number in the following series of numbers?

73, 62, 54, 49, 47, ?

A) 45

B) 44

C) 47

D) 48

WORKSPACE

Question 9

What is the next number in the following series of numbers?

6, 11, 9, 16, 12, 21, ?

A) 14

B) 15

C) 16

D) 18

WORKSPACE

Question 10

What is the next number in the following series of numbers?

4, 10, 7, 14, 20, 17, 24, 30, ?

A) 25

B) 27

C) 28

D) 31

WORKSPACE

Question 11

Consider the formula. If B = 3 and C = 5, what is the value of A?

$$(4A - (B + 2C)) + 3 = AB - C$$

A) 2

B) 3

C) 4

D) 5

WORKSPACE

Question 12

Consider the formula. If A = 4 and D = 6, what is the value of B?

$$D(B + 2) = 44 - (B + A(B - 3))$$

A) 1

B) 2

C) 3

D) 4

WORKSPACE

Question 13

Simplify the following expression to its simplest form, when B = 8 and C = 2.

$$2(A^3 + A)/A - (A^4 - A^2)/A^2 + (B + C)(B - C) - A^2$$

A) 63

B) 60

C) 62

D) $A^2 + 63$

WORKSPACE

Question 14

Consider the formula. Which statement is true?

$$A = 10 - 2B$$

A) If A is negative, then B is greater than 5.

B) If A is greater than 10, then B is positive.

C) If B is negative, then A is negative.

D) If B is greater than 5, then A is positive.

WORKSPACE

Question 15

Consider the formula. Which statement is true?

$$x = 0.25y - 4$$

A) When the value of y is greater than 16, x is positive.

B) When the value of y is less than 16, x is positive.

C) When the value of y is greater than 16, x is negative.

D) When the value of y is less than 16, x is zero.

WORKSPACE

Question 16

Consider the formula. Which statement is true?

$$H = 2M + 2$$

A) The difference between the values of H and M changes.

B) The difference between the values of H and M is constant.

C) For positive H values, $M > 0$.

D) For negative M values, $H < 0$.

WORKSPACE

Question 17

Consider the formula. Which statement is true?

$$3x = 6y + 12$$

A) When the value of y is less than −2, x is positive.

B) When the value of x is less than 4, y is positive.

C) When the value of x is greater than 10, y is negative.

D) When the value of y is greater than −2, x is positive.

WORKSPACE

Question 18

Consider the formula. Which statement is true?

$$(1/3)x = 5 - (1/2)y$$

A) When the value of y is greater than 10, x is positive.

B) When the value of y is less than 10, x is positive.

C) When the value of x is less than 15, y is negative.

D) When the value of x is greater than 15, y is positive.

WORKSPACE

Question 19

Consider the formula. Which statement is true?

$$H = M - 4$$

A) $H + M > 0$

B) $H + M > 2M$

C) $M - H < 0$

D) $(M - H) > (H - M)$

WORKSPACE

Question 20

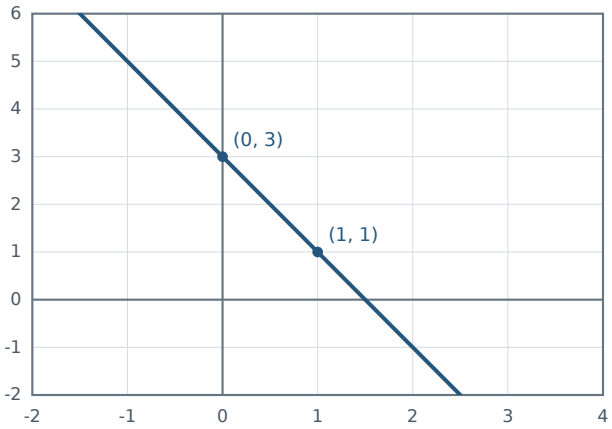
Which choice represents the same relationship as the given function?

$y = 2x - 3$

A)

x	y
-1	-5
0	-3
1	-1
2	1
3	3

B)



C)

x	y
-1	-1
0	-3
1	-5
2	-7
3	-9

D) y is equal to x minus 3.

WORKSPACE

Question 21

Which choice represents the same relationship as the information below?

y is equivalent to twelve times the value of x divided by three.

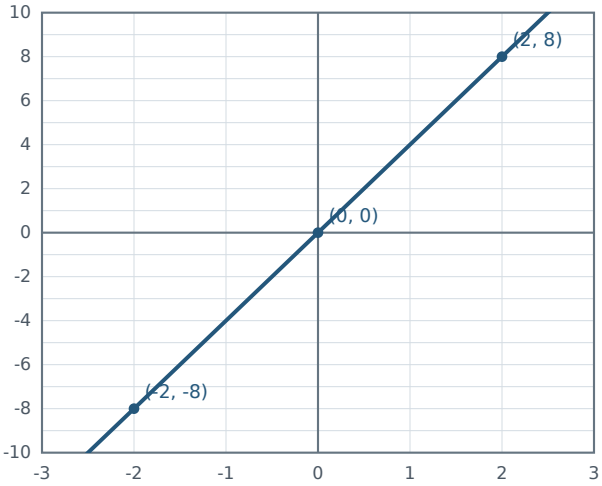
A)

x	y
0	0
1	4
2	8
3	12
4	16

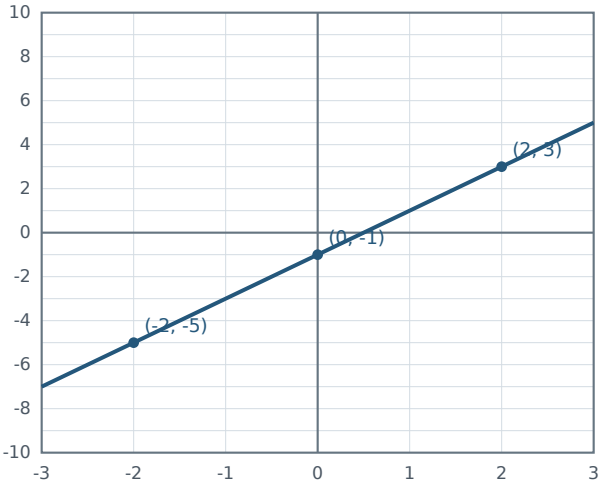
B)

x	y
0	-1
1	1
2	3
3	5
4	7

C)



D)



WORKSPACE

Question 22

Consider the table. Which choice represents the same relationship as the given table?

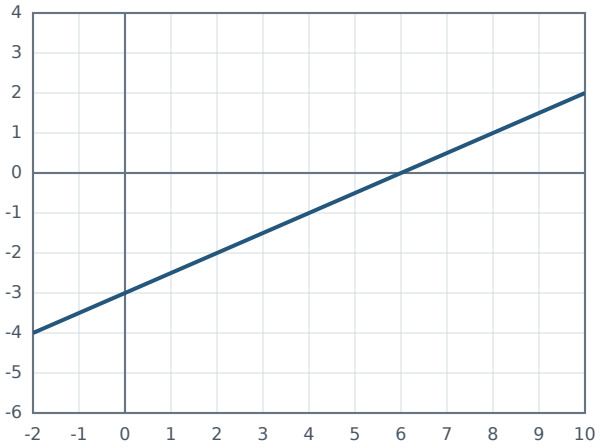
x	y
0	-3
2	-2
4	-1
6	0
8	1

A) y is equal to x − 3.

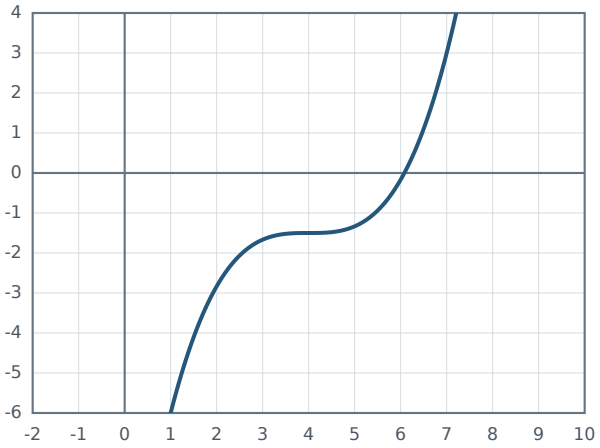
B)

x	y
0	-3
4	-2
8	-1
12	0
1	0

C)



D)

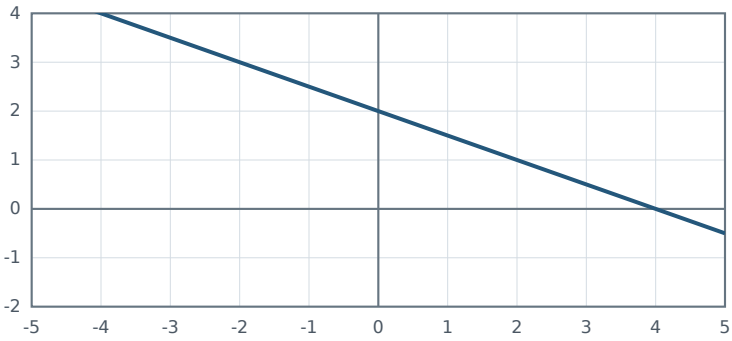


WORKSPACE

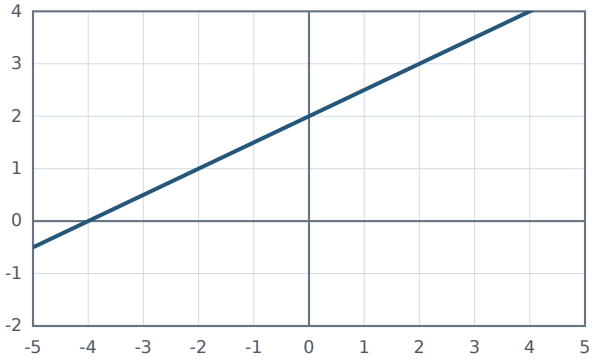
Question 23

Consider the graph of the function y = mx + b. Which option is true regarding this graph?

Given graph



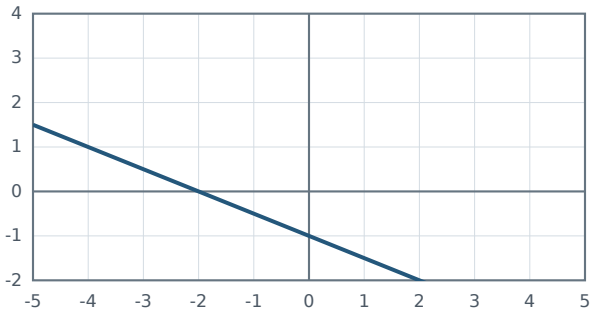
A)



B) Function B, y = −(1/2)x + 2, has the same slope and the same intercept point with the y-axis.

C) Function C, y = −2x + 2, has the same slope and the same intercept point with the y-axis.

D)



WORKSPACE

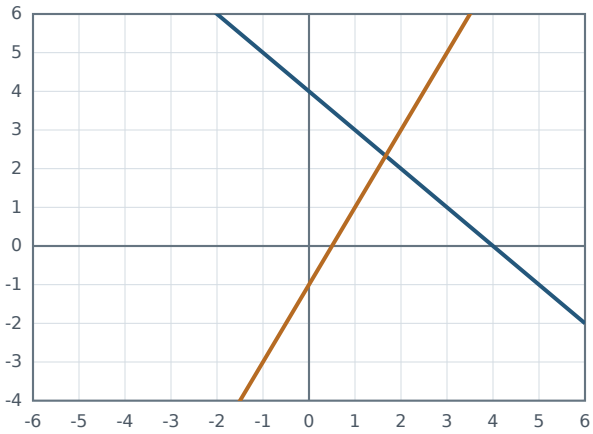
Question 24

Which graph represents the following two linear equations?

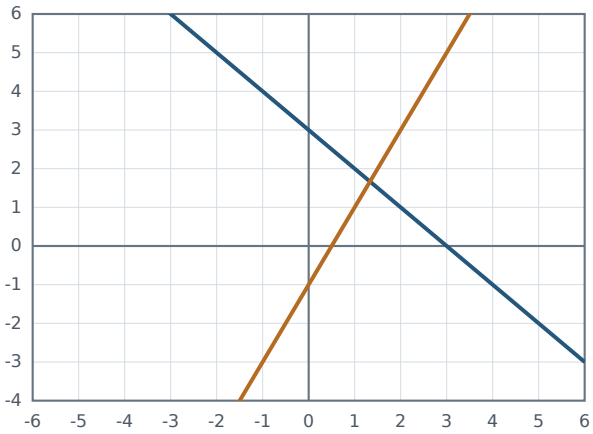
$x + y = 4$

$2x - y = 1$

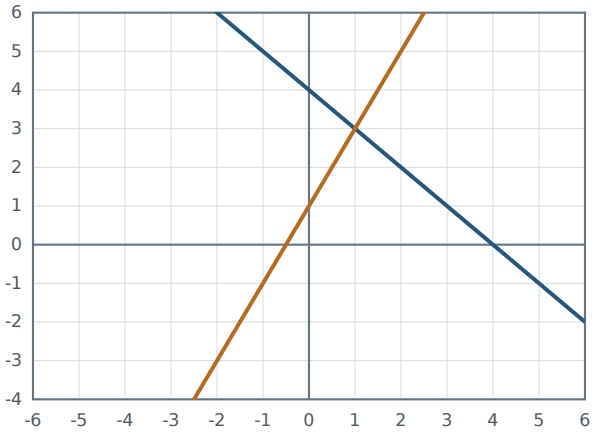
A)



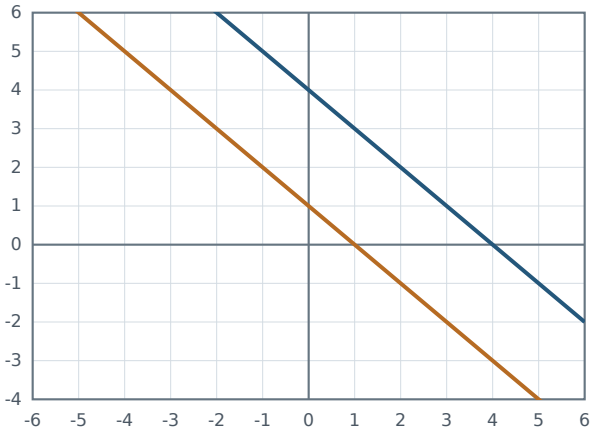
B)



C)



D)



WORKSPACE

Question 25

Solve the proportion. Which value of m is correct?

$$(m - 4)/(16 - m) = 3/9$$

A) 6

B) 7

C) 10

D) 12

WORKSPACE

Question 26

Evaluate the expression for x = 6 and y = (5x − 3)/6.

(3x − 2)/(2y + 1) + (2y + 1)/(3x − 2) = ?

A) 2.125

B) 2.225

C) 2.250

D) 2.300

WORKSPACE

Question 27

Evaluate the formula:

$$Q = 1/r$$

$$r = n/(2\pi)$$

$$Q = ?$$

A) $2\pi/n$

B) $n/(2\pi)$

C) r/n

D) $r/(2Q)$

WORKSPACE

Question 28

Solve the following system of equations. Which ordered pair (x, y) is the solution?

$$2x + 3y = 13$$
$$4x - y = 5$$

A) (2, 3)

B) (1, 2)

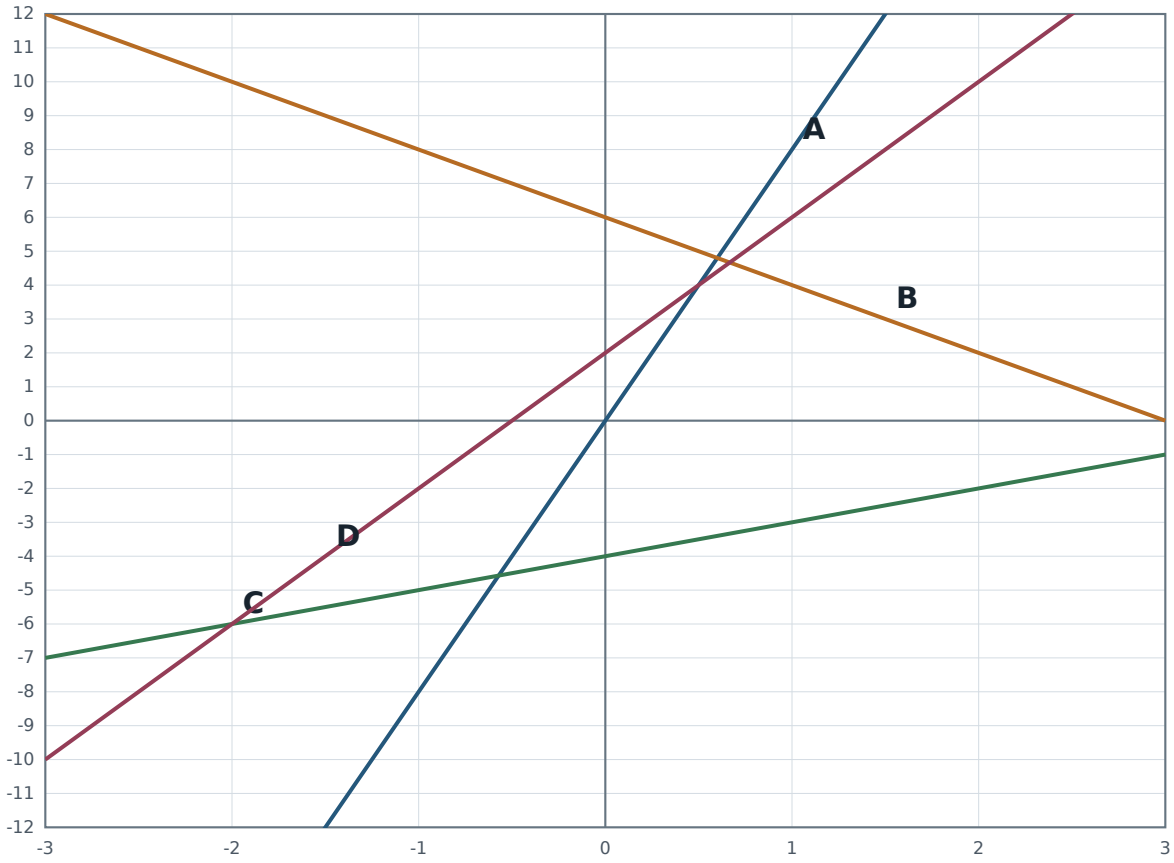
C) (2, −1)

D) (−1, 5)

WORKSPACE

Question 29

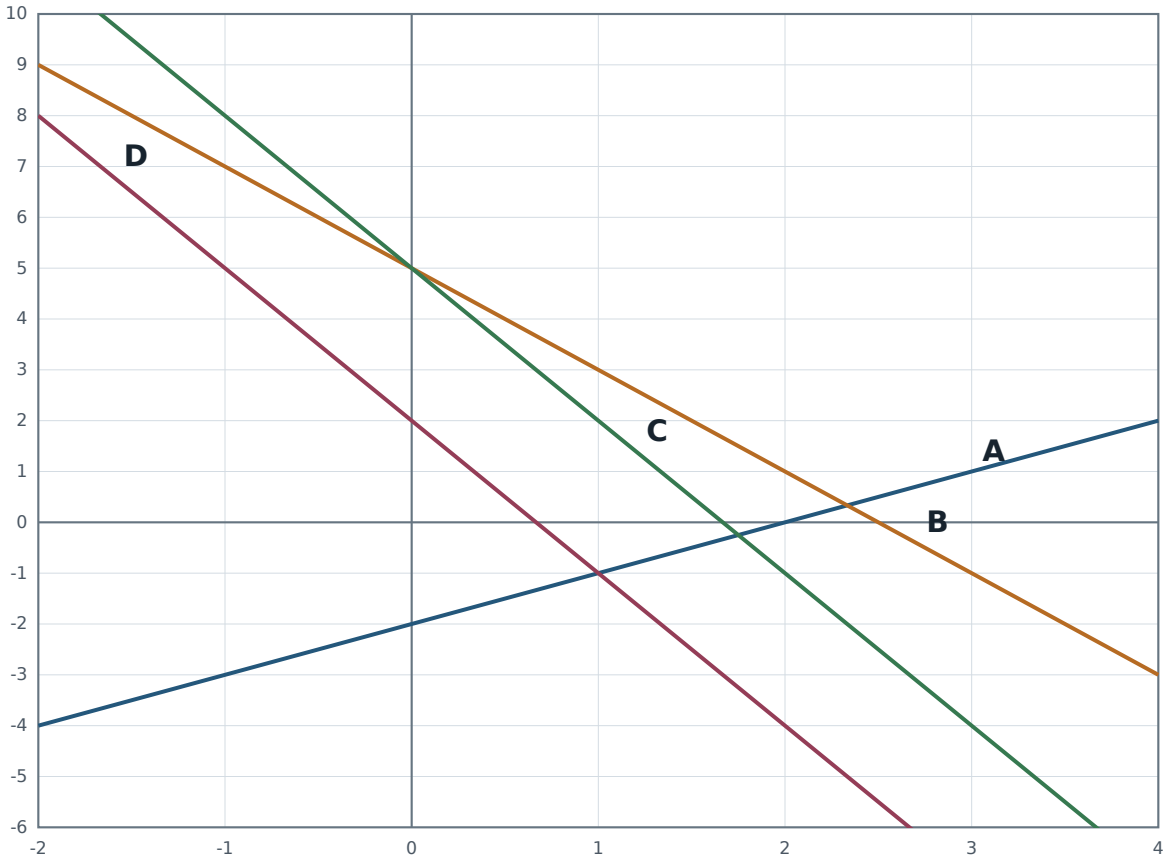
Which graph represents the function y = 8x?



WORKSPACE

Question 30

Which graph represents the function y = −3x + 5?



WORKSPACE

Question 31

Find the unknown variable M, given p = 24, q = 10, r = 4, s = 1.

$$p/(q - r) + p(s - r)/(q - s) = M$$

A) −4

B) −2

C) 0

D) 2

WORKSPACE

Question 32

Isolate C from the formula:

$$XY = A(B + C)/D + E/F$$

$$C = ?$$

A) $D(XY - E/F)/A - B$

B) $A/D(XY - E/F) - B$

C) $D(XY + E/F)/A - B$

D) $A(B + C)/D - XY + E/F$

WORKSPACE

Question 33

Isolate p_1 from the formula:

$$M = p_1(p_2 - 3)/4$$

$$p_1 = ?$$

A) $(4M)/p_2 - 3$

B) $(M(p_2 - 3))/4$

C) $(p_2 - 3)/(4M)$

D) $(4M/p_2) - 3$

WORKSPACE